

NASA's DART Smash-Up Changed an Asteroid's Path Around the Sun!

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What if you could throw a baseball at a giant space rock and change its path around the sun? That's basically what NASA did with its DART spacecraft and it worked even better than expected!

Four years ago, NASA slammed DART into a small **asteroid** called Dimorphos, which **orbits** a bigger one named Didymos. The goal was to see if we could push an **asteroid** off course, just in case one ever threatens Earth. Right after the crash, scientists saw that Dimorphos moved closer to Didymos. But now, a new study shows something even cooler: the whole **asteroid** pair's **orbit** around the sun changed!

How much? Their speed slowed by just 10 **micrometers** per second. That's slower than a snail, but in space, it's a huge deal. "Tweaking the motions of space rocks millions of kilometers from Earth is a huge deal," said researcher Rahil Makadia. To measure this, astronomers watched the **asteroids** pass in front of distant stars, like tiny **eclipses**. **Amateur astronomers** even drove two days into the Australian outback to help!

The European Space Agency's Hera spacecraft will visit the **asteroids** later this year to confirm the new **orbit**. This isn't just cool science it's practice for protecting our planet. If a dangerous **asteroid** ever heads our way, we now know we can change its path. That's **planetary defense** in action!

Word Treasure

asteroid -- A rocky object in space, smaller than a planet, that orbits the sun.

orbit -- The path one object takes as it moves around another object in space.

micrometers -- A unit of length -- one micrometer is a millionth of a meter, as thin as a strand of spider silk.

eclipses -- Moments when one object in space passes in front of another and blocks its light.

amateur astronomers -- People who love watching stars and space as a hobby, not as their job.

planetary defense -- The science and plans for protecting Earth from dangerous asteroids or comets.

Background you need

DART stands for Double Asteroid Redirection Test -- the first mission to try changing an asteroid's motion by crashing into it.

Dimorphos and Didymos are part of a double-asteroid system; hitting the smaller one was safer because scientists could measure the effect without creating new hazards.

The dinosaurs died out 66 million years ago because a large asteroid hit Earth -- that's why testing planetary defense is so important today.

Break it down

WHO: NASA scientists, researchers like Rahil Makadia, amateur astronomers, and the European Space Agency's Hera team.

WHAT: The DART impact not only moved Dimorphos closer to Didymos but also slowed the whole pair's orbit around the sun.

WHERE: Deep space, 11 million kilometers from Earth, and observations made from the Australian outback.

WHY: To prove we can change an asteroid's path and build a real system for planetary defense.

Why it matters

One day, knowing how to push an asteroid could save the whole planet, just like a fire drill at school prepares you for an emergency that might never happen.

Test what you remember

1. What did NASA's DART spacecraft do to the asteroid Dimorphos?
 - (A) Flew past it and took pictures
 - (B) Landed on it and collected rocks
 - (C) Slammed into it to push it off course
 - (D) Orbited around it for a year
2. What new discovery did scientists make about the asteroid pair?
 - (A) They split into three pieces
 - (B) Their whole orbit around the sun changed
 - (C) They disappeared from view
 - (D) They grew in size
3. By how much did the asteroids' speed slow down?
 - (A) 10 meters per second
 - (B) 10 micrometers per second
 - (C) 10 kilometers per hour
 - (D) 10 millimeters per day
4. How did astronomers measure the tiny change in orbit?
 - (A) With giant telescopes on mountains
 - (B) By watching the asteroids pass in front of distant stars
 - (C) By sending another spacecraft right away
 - (D) With super-powered lasers from Earth
5. Which space agency's Hera spacecraft will visit the asteroids later this year?
 - (A) NASA
 - (B) European Space Agency
 - (C) Japan Aerospace Exploration Agency
 - (D) Russian Space Agency
6. What is the main reason scientists are excited about this test?
 - (A) It proves we can grow plants on asteroids
 - (B) It shows we can defend Earth from dangerous space rocks
 - (C) It teaches us how to build space cities
 - (D) It finds new types of rock

Answer key (try the quiz first!): 1: C 2: B 3: B 4: B 5: B 6: B

Questions to think about

- 1. Success Celebration:** The bigger-than-expected change shows that with enough warning, we could deflect a dangerous asteroid, making Earth safer.
- 2. Confirmation Needed:** We still need Hera's close-up visit to confirm the long-term orbit change and see what the crash did to Dimorphos's surface.
- 3. Community Science:** Ordinary amateur astronomers drove far into the desert to help gather important data, proving you don't need a giant lab to contribute to space science.

MY THOUGHTS (at least 20 words):